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LẬP TRÌNH ỨNG DỤNG C# NET CORE

<Tên đề tài của sinh viên>

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PHIẾU NHẬN XÉT VÀ ĐÁNH GIÁ CỦA GIẢNG VIÊN HƯỚNG DẪN

MỤC LỤC

LỜI CẢM ƠN & LỜI CAM ĐOAN 2
PHIẾU NHẬN XÉT VÀ ĐÁNH GIÁ CỦA GIẢNG VIÊN HƯỚNG DẪN 2
CHƯƠNG 1: TỔNG QUAN ĐỀ TÀI VÀ CƠ SỞ LÝ THUYẾT NỀN TẢNG 13
1.1. Tổng quan đề tài nghiên cứu 13
1.1.1. Bối cảnh chuyển đổi số trong quản lý bán lẻ và chuỗi cửa hàng tiện lợi / siêu thị mini 13
1.1.2. Lý do chọn đề tài và tính cấp thiết của việc xây dựng hệ thống quản lý bán lẻ 13
1.1.3. Mục tiêu nghiên cứu và chuẩn đầu ra của đồ án môn học 14
1.1.4. Đối tượng và phạm vi nghiên cứu của đề tài 15
1.2. Cơ sở lý thuyết về nền tảng Microsoft .NET Core và ngôn ngữ C# 16
1.2.1. Quá trình tiến hóa từ .NET Framework sang Modern .NET (.NET Core / .NET 8) 16
1.2.2. Cơ chế thực thi mã nguồn: Mã trung gian (CIL), Trình biên dịch JIT và Bộ máy thực thi CLR 17
1.2.3. Cơ chế quản lý bộ nhớ tự động: Bộ nhớ Stack, Heap và Trình thu gom rác (Garbage Collector) 17
1.2.4. Các cú pháp hiện đại trong C#: Auto-properties, Generics, Nullable types và Lập trình bất đồng bộ (async/await) 18
1.3. Lý thuyết về Kiến trúc hướng dịch vụ (SOA) và chuẩn RESTful Web API 19
1.3.1. So sánh kiến trúc nguyên khối (Monolithic) và Kiến trúc hướng dịch vụ (Service-Oriented Architecture - SOA) 19
1.3.2. Nguyên lý thiết kế RESTful: Định danh tài nguyên (URI), quy chuẩn danh từ số nhiều và tính phi trạng thái (Stateless) 20
1.3.3. Giao thức truyền tải HTTP: Phương thức (GET, POST, PUT, DELETE) và Quy chuẩn mã phản hồi (HTTP Status Codes) 21
1.3.4. Định dạng trao đổi dữ liệu JSON: Kỹ thuật tuần tự hóa (Serialization) và giải tuần tự hóa (Deserialization) 21
1.3.5. Kiến trúc xử lý yêu cầu trong ASP.NET Core: Bộ chứa phụ thuộc (DI Container) và Chuỗi xử lý trung gian (Middleware Pipeline) 22
1.4. Cơ sở lý thuyết về Cơ sở dữ liệu và Tầng truy xuất dữ liệu 23
1.4.1. Hệ quản trị cơ sở dữ liệu quan hệ (RDBMS) và chuẩn hóa dữ liệu 3NF trên Microsoft SQL Server 23
1.4.2. Khái niệm Object-Relational Mapping (ORM) và cơ chế vận hành của Entity Framework Core (EF Core) 23
1.4.3. Phương pháp tiếp cận Code-First, DbContext và Quản lý tiến hóa lược đồ qua EF Core Migrations 23
1.4.4. Ngôn ngữ truy vấn tích hợp LINQ (Language Integrated Query) và cơ chế Deferred Execution 23
1.4.5. Công nghệ ADO.NET thuần: Kiến trúc hướng kết nối/ngắt kết nối, Stored Procedure và Tiêu chuẩn giao dịch ACID (SqlTransaction) 23

1.5. Lý thuyết về Cơ chế Bảo mật và Xác thực không trạng thái (JWT) 23
1.5.1. So sánh cơ chế xác thực dựa trên Session truyền thống và Xác thực không trạng thái (Stateless Authentication) 23
1.5.2. Cấu trúc chuẩn của chuỗi JSON Web Token: Header, Payload (Claims), và Signature 24
1.5.3. Thuật toán ký mã hóa đối xứng (HMAC SHA-256) và Quy trình ủy quyền truy cập theo vai trò (Role-Based Access Control - RBAC) 25
1.6. Kiến trúc Desktop Client trên nền tảng Windows Forms 26
1.6.1. Nguyên lý giao tiếp giữa ứng dụng máy trạm và Backend API thông qua lớp HttpClient 26
1.6.2. Kỹ thuật quản trị tài nguyên mạng: Khắc phục sự cố cạn kiệt cổng kết nối (Socket Exhaustion) 27
1.6.3. Kiến trúc giao diện bàn làm việc đơn (Single-Form Shell Workspace) và cơ chế nhúng Form động 27
CHƯƠNG 2: PHÂN TÍCH YÊU CẦU VÀ THIẾT KẾ HỆ THỐNG 29
2.1. Phân tích yêu cầu chức năng và luồng nghiệp vụ hệ thống siêu thị mini 29
2.1.1. Luồng phân quyền và xác thực người dùng (Quản trị viên - Admin, Thu ngân - Cashier, Thủ kho - Warehouse) 29
2.1.2. Luồng nghiệp vụ quản lý danh mục nhóm hàng và mặt hàng tồn kho 29
2.1.3. Luồng nghiệp vụ quầy thu ngân bán lẻ (POS): Tra cứu mã vạch, tính tiền, trừ kho tự động và tích điểm 30
2.1.4. Luồng nghiệp vụ báo cáo thống kê doanh thu và hiệu suất ca trực 31
2.2. Thiết kế mô hình kiến trúc Client - Server tổng thể 32
2.2.1. Sơ đồ kiến trúc 3 lớp: Desktop WinForms Client ASP.NET Core Web API SQL Server Database 32
2.2.2. Sơ đồ tuần hoàn dòng chảy dữ liệu (Data Flow Diagram) trong hệ thống 33
2.3. Thiết kế Cơ sở dữ liệu quan hệ (Database Schema) 33
2.3.1. Sơ đồ thực thể liên kết (ERD) hoàn chỉnh gồm 6 bảng thực thể nghiệp vụ cốt lõi 33
2.3.2. Phân hệ tài khoản và phân quyền: Chi tiết bảng Users 33
2.3.3. Phân hệ hàng hóa và nhóm hàng: Chi tiết bảng Categories và Products 33
2.3.4. Phân hệ khách hàng thành viên: Chi tiết bảng Customers 33
2.3.5. Phân hệ hóa đơn và giao dịch bán lẻ: Chi tiết bảng Orders và OrderDetails 33
2.3.6. Bảng tổng hợp các ràng buộc toàn vẹn dữ liệu: Khóa chính (PK), Khóa ngoại (FK), Ràng buộc Duy nhất (Unique) và Giá trị mặc định 33
2.4. Thiết kế danh mục và đặc tả chuẩn RESTful Web API 34
2.4.1. Quy chuẩn định dạng đường dẫn Endpoint, Header và cấu trúc gói tin JSON 34
2.4.2. Phân hệ API Xác thực và Cấp phát Token (/api/auth) 35
2.4.3. Phân hệ API Quản lý Nhóm hàng (/api/categories) 35
2.4.4. Phân hệ API Quản lý Sản phẩm, Tra cứu Barcode và Phân trang (/api/products, /api/ado-products) 36

2.4.5. Phân hệ API Giao dịch Thanh toán POS (/api/checkout) 36
2.4.6. Phân hệ API Báo cáo Doanh thu Stored Procedure (/api/reports) 36
CHƯƠNG 3: TRIỂN KHAI PHẦN BACKEND (ASP.NET CORE WEB API) 38
3.1. Cấu trúc Solution và phân tách các tầng kiến trúc 38
3.1.1. Cấu trúc thư mục dự án MiniSupermarket.API (Controllers, Data, Models, DTOs) 38
3.1.2. Thiết lập chuỗi kết nối và thông số bảo mật an toàn trong appsettings.json 38
3.1.3. Cấu hình tệp khởi động Program.cs: Đăng ký DI Container, Kích hoạt Middleware Pipeline và cấu hình CORS 39
3.2. Hiện thực tầng dữ liệu với Entity Framework Core Code-First 39
3.2.1. Định nghĩa các thực thể C# (Model Classes) kết hợp thuộc tính Data Annotations 39
3.2.2. Xây dựng lớp MiniSupermarketDbContext và cấu hình liên kết bảng qua Fluent API 40
3.2.3. Thực thi EF Core Migrations và khởi tạo bộ dữ liệu mẫu ban đầu (Data Seeding) 40
3.2.4. Triển khai các thao tác CRUD danh mục và hàng hóa qua Entity Framework Core 40
3.2.5. Tối ưu hóa truy vấn bảng liên quan bằng Eager Loading (.Include()) và giải quyết vòng lặp tham chiếu JSON 40
3.2.6. Thuật toán phân trang tầng máy chủ (Server-side Pagination) với LINQ .Skip() / .Take() kết hợp tối ưu .AsNoTracking() 40
3.3. Hiện thực phân hệ xác thực và phân quyền bảo mật (JWT & RBAC) 40
3.3.1. Xây dựng AuthController xử lý nghiệp vụ mã hóa mật khẩu và cấp phát JWT Bearer Token 40
3.3.2. Cấu hình dịch vụ xác thực AddJwtBearer và thuật toán khóa ký đối xứng SymmetricSecurityKey trong Program.cs 41
3.3.3. Thiết lập phân quyền truy cập đầu cuối sử dụng thuộc tính [Authorize] và [Authorize(Roles = "...")] 42
3.4. Hiện thực kỹ thuật ADO.NET nâng cao và Quản trị Giao dịch POS Checkout 43
3.4.1. Xây dựng phân hệ tra cứu mã vạch siêu tốc bằng SqlConnection, SqlCommand và luồng đọc SqlDataReader 43
3.4.2. Phòng chống triệt để lỗ hổng tấn công SQL Injection thông qua kỹ thuật tham số hóa SqlParameter 44
3.4.3. Lập trình và thực thi Stored Procedure sp_GetRevenueByCashier kết xuất báo cáo doanh thu ca làm việc 45
3.4.4. Đảm bảo toàn vẹn chuẩn ACID: Bọc kín chuỗi 4 thao tác thanh toán trong SqlTransaction (Commit / Rollback) 46
3.5. Tài liệu hóa và chuẩn hóa kiểm thử API bằng Swagger / OpenAPI UI 46
3.5.1. Tích hợp thư viện Swashbuckle và cấu hình giao diện SwaggerGen 46

3.5.2. Cấu hình nút xác thực ổ khóa "Authorize" để truyền Bearer Token trực tiếp trên giao diện Swagger 47
CHƯƠNG 4: TRIỂN KHAI PHẦN FRONTEND CLIENT (WINDOWS FORMS POS APP) 47
4.1. Kiến trúc tổng thể và giao tiếp dịch vụ phía WinForms Client 47
4.1.1. Cấu trúc Solution dự án MiniSupermarket.WinForms 47
4.1.2. Xây dựng lớp dịch vụ mạng tập trung ApiClientService.cs quản lý phiên bản static readonly HttpClient 48
4.1.3. Cơ chế tự động đính kèm Bearer Token vào Header trong các yêu cầu bất đồng bộ qua System.Net.Http.Json 48
4.1.4. Lớp tĩnh SessionManager.cs: Quản lý phiên làm việc, lưu trữ định danh tài khoản và vai trò người dùng 48
4.2. Hiện thực kiến trúc bàn làm việc Shell Workspace (Single-Form Shell) 49
4.2.1. Phân chia bố cục phân vùng giao diện: Thanh điều hướng bên trái (Sidebar), Thanh tiêu đề (Top Header) và Vùng hiển thị (Main Workspace) 49
4.2.2. Kỹ thuật nhúng Form con động (TopLevel = false, Dock = Fill) vào Panel trung tâm, ngăn chặn việc mở cửa sổ độc lập phân mảnh 50
4.2.3. Cơ chế điều khiển phân quyền giao diện: Tự động ẩn/hiện nút chức năng Sidebar theo quyền hạn đăng nhập thực tế (RBAC) 51
4.3. Xây dựng các phân hệ màn hình nghiệp vụ Desktop 52
4.3.1. Màn hình Xác thực đăng nhập (FormLogin.cs): Giao tiếp API xác thực, giải mã phản hồi và nạp phiên làm việc 52
4.3.2. Màn hình Quầy Bán lẻ POS (FormPOS.cs): Bắt sự kiện quét mã Barcode, quản lý bảng giỏ hàng, tự động tính tổng tiền, tính tiền thừa và kích hoạt Checkout qua API 52
4.3.3. Màn hình Quản lý Sản phẩm & Tồn kho (FormProductManagement.cs): Nạp danh mục ComboBox, hiển thị lưới dữ liệu DataGridView và cụm nút điều hướng phân trang (Next/Prev) 53
4.3.4. Màn hình Quản lý Danh mục Nhóm hàng (FormCategoryManagement.cs): Thao tác Thêm, Sửa, Xóa và Tìm kiếm trực tiếp 54
4.3.5. Màn hình Dashboard Báo cáo Doanh thu (FormReports.cs): Trực quan hóa số liệu ca trực bằng bảng biểu và điều khiển biểu đồ đồ họa 55
4.4. Xử lý ngoại lệ và nâng cao trải nghiệm người dùng trên Desktop Client 56
4.4.1. Bọc kín các tiến trình mạng bằng cấu trúc xử lý lỗi ngoại lệ (try ... catch), chống treo hoặc sập ứng dụng 56
4.4.2. Xử lý thông báo phản hồi thân thiện: Phân biệt các trạng thái lỗi mạng, hết hạn phiên làm việc (401) hoặc từ chối truy cập (403) 56
CHƯƠNG 5: KIỂM THỬ HỆ THỐNG VÀ ĐÁNH GIÁ KẾT QUẢ ĐỒ ÁN 57
5.1. Môi trường và Kịch bản kiểm thử hệ thống 57
5.1.1. Cấu hình môi trường thử nghiệm: Phần cứng máy trạm, Máy chủ Web API và Cơ sở dữ liệu SQL Server 57
5.1.2. Bộ dữ liệu kiểm thử chuẩn hóa: 15 tài khoản nhân sự mẫu phân quyền và 15 mặt hàng kiểm thử tồn kho 58

5.2. Kiểm thử phân hệ Backend Web API (Swagger UI & Postman) 59
5.2.1. Kiểm thử tính năng cấp Token JWT và gọi API công khai (200 OK / 201 Created) 59
5.2.2. Kiểm thử cơ chế bảo vệ phân quyền: Kiểm tra mã trạng thái 401 Unauthorized và 403 Forbidden khi truy cập vượt quyền 60
5.2.3. Kiểm thử phân trang máy chủ: Đánh giá thời gian phản hồi khi trích xuất dữ liệu với các bộ tham số pageNumber và pageSize 60
5.3. Kiểm thử nghiệp vụ Giao dịch toàn vẹn dữ liệu (ACID Transaction Testing) 61
5.3.1. Kịch bản 1 (Thanh toán thành công - Commit): Kiểm tra đối soát đồng bộ dữ liệu tại cả 4 bảng (Orders, OrderDetails, Products, Customers) trên SSMS 61
5.3.2. Kịch bản 2 (Hoàn tác tự động - Rollback): Cố tình đặt mua mặt hàng vượt quá số lượng tồn kho thực tế, xác nhận hệ thống hủy giao dịch an toàn và không phát sinh hóa đơn rác 62
5.4. Kiểm thử tích hợp toàn diện trên ứng dụng WinForms POS Client 63
5.4.1. Thử nghiệm luồng bán hàng hoàn chỉnh không dùng chuột tại quầy POS thông qua máy quét Barcode và phím tắt 63
5.4.2. Thử nghiệm đồng bộ dữ liệu thời gian thực: Thao tác Thêm/Sửa/Xóa hàng hóa từ Client và kiểm tra sự thay đổi trực tiếp dưới SQL Server 64
5.5. Đánh giá ưu điểm và nhược điểm của giải pháp thiết kế 65
5.5.1. Những ưu điểm nổi bật về mặt kiến trúc, tính bảo mật và hiệu năng vận hành 65
5.5.2. Những điểm hạn chế còn tồn tại của hệ thống 65
CHƯƠNG 6: KẾT LUẬN VÀ HƯỚNG PHÁT TRIỂN NÂNG CẤP 66
6.1. Kết quả đạt được của đồ án đối với sinh viên 66
6.1.1. Mức độ hoàn thành các chức năng so với yêu cầu và đề cương môn học 66
6.1.2. Những kiến thức học thuật và kỹ năng kỹ thuật phần mềm tích lũy được qua đồ án (.NET Core, RESTful, EF Core, ADO.NET, WinForms) 67
6.2. Hướng phát triển và nâng cấp hệ thống trong tương lai 68
6.2.1. Tích hợp thanh toán trực tuyến thông qua chuẩn mã VietQR động hoặc Cổng thanh toán điện tử (VNPay, MoMo) 68
6.2.2. Mở rộng ứng dụng Web dành cho khách hàng tự đặt hàng (E-Commerce) và Ứng dụng Di động dành cho nhân viên kho kiểm kê hàng hóa qua camera 69
6.2.3. Tích hợp thuật toán trí tuệ nhân tạo (AI/Machine Learning) dự báo nhu cầu nhập hàng và gợi ý sản phẩm bán kèm tại quầy thu ngân 70
TÀI LIỆU THAM KHẢO (Sách giáo trình chuyên ngành, Tài liệu học thuật Microsoft Learn, Tiêu chuẩn mã nguồn .NET Core) 71
PHỤ LỤC: HƯỚNG DẪN CÀI ĐẶT & VẬN HÀNH HỆ THỐNG 71
Phụ lục A: Hướng dẫn cấu hình kết nối SQL Server và các lệnh thực thi EF Core Migrations 71
Phụ lục B: Danh sách bảng tài khoản người dùng mẫu và phân quyền truy cập hệ thống 72
Phụ lục C: Bảng thống kê chi tiết mã nguồn các Controller và Form giao diện chính 73

PHỤ LỤC (Đường link GitHub, Tài khoản/Mật khẩu demo, Hướng dẫn cài đặt nhanh) 75

DANH MỤC HÌNH ẢNH, SƠ ĐỒ

- **DANH MỤC THUẬT NGỮ & TỪ VIẾT TẮT *(SOA, RESTful, JWT, ORM, DTO, ACID, POS, EF Core, RDBMS)***

- **DANH MỤC HÌNH ẢNH & SƠ ĐỒ**

- **DANH MỤC BẢNG BIỂU**

CHƯƠNG 1: TỔNG QUAN ĐỀ TÀI VÀ CƠ SỞ LÝ THUYẾT NỀN TẢNG

1.1. Tổng quan đề tài nghiên cứu

1.1.1. Bối cảnh chuyển đổi số trong quản lý bán lẻ và chuỗi cửa hàng tiện lợi / siêu thị mini

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1.1.2. Lý do chọn đề tài và tính cấp thiết của việc xây dựng hệ thống quản lý bán lẻ

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1.1.3. Mục tiêu nghiên cứu và chuẩn đầu ra của đồ án môn học

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1.1.4. Đối tượng và phạm vi nghiên cứu của đề tài

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1.2. Cơ sở lý thuyết về nền tảng Microsoft .NET Core và ngôn ngữ C#

1.2.1. Quá trình tiến hóa từ .NET Framework sang Modern .NET (.NET Core / .NET 8)

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1.2.2. Cơ chế thực thi mã nguồn: Mã trung gian (CIL), Trình biên dịch JIT và Bộ máy thực thi CLR

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1.2.3. Cơ chế quản lý bộ nhớ tự động: Bộ nhớ Stack, Heap và Trình thu gom rác (Garbage Collector)

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1.2.4. Các cú pháp hiện đại trong C#: Auto-properties, Generics, Nullable types và Lập trình bất đồng bộ (async/await)

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1.3. Lý thuyết về Kiến trúc hướng dịch vụ (SOA) và chuẩn RESTful Web API

1.3.1. So sánh kiến trúc nguyên khối (Monolithic) và Kiến trúc hướng dịch vụ (Service-Oriented Architecture - SOA)

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1.3.2. Nguyên lý thiết kế RESTful: Định danh tài nguyên (URI), quy chuẩn danh từ số nhiều và tính phi trạng thái (Stateless)

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1.3.3. Giao thức truyền tải HTTP: Phương thức (GET, POST, PUT, DELETE) và Quy chuẩn mã phản hồi (HTTP Status Codes)

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1.3.4. Định dạng trao đổi dữ liệu JSON: Kỹ thuật tuần tự hóa (Serialization) và giải tuần tự hóa (Deserialization)

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1.3.5. Kiến trúc xử lý yêu cầu trong ASP.NET Core: Bộ chứa phụ thuộc (DI Container) và Chuỗi xử lý trung gian (Middleware Pipeline)

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1.4. Cơ sở lý thuyết về Cơ sở dữ liệu và Tầng truy xuất dữ liệu

1.4.1. Hệ quản trị cơ sở dữ liệu quan hệ (RDBMS) và chuẩn hóa dữ liệu 3NF trên Microsoft SQL Server

1.4.2. Khái niệm Object-Relational Mapping (ORM) và cơ chế vận hành của Entity Framework Core (EF Core)

1.4.3. Phương pháp tiếp cận Code-First, DbContext và Quản lý tiến hóa lược đồ qua EF Core Migrations

1.4.4. Ngôn ngữ truy vấn tích hợp LINQ (Language Integrated Query) và cơ chế Deferred Execution

1.4.5. Công nghệ ADO.NET thuần: Kiến trúc hướng kết nối/ngắt kết nối, Stored Procedure và Tiêu chuẩn giao dịch ACID (SqlTransaction)

1.5. Lý thuyết về Cơ chế Bảo mật và Xác thực không trạng thái (JWT)

1.5.1. So sánh cơ chế xác thực dựa trên Session truyền thống và Xác thực không trạng thái (Stateless Authentication)

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1.5.2. Cấu trúc chuẩn của chuỗi JSON Web Token: Header, Payload (Claims), và Signature

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1.5.3. Thuật toán ký mã hóa đối xứng (HMAC SHA-256) và Quy trình ủy quyền truy cập theo vai trò (Role-Based Access Control - RBAC)

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1.6. Kiến trúc Desktop Client trên nền tảng Windows Forms

1.6.1. Nguyên lý giao tiếp giữa ứng dụng máy trạm và Backend API thông qua lớp HttpClient

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1.6.2. Kỹ thuật quản trị tài nguyên mạng: Khắc phục sự cố cạn kiệt cổng kết nối (Socket Exhaustion)

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1.6.3. Kiến trúc giao diện bàn làm việc đơn (Single-Form Shell Workspace) và cơ chế nhúng Form động

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CHƯƠNG 2: PHÂN TÍCH YÊU CẦU VÀ THIẾT KẾ HỆ THỐNG

2.1. Phân tích yêu cầu chức năng và luồng nghiệp vụ hệ thống siêu thị mini

2.1.1. Luồng phân quyền và xác thực người dùng (Quản trị viên - Admin, Thu ngân - Cashier, Thủ kho - Warehouse)

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2.1.2. Luồng nghiệp vụ quản lý danh mục nhóm hàng và mặt hàng tồn kho

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2.1.3. Luồng nghiệp vụ quầy thu ngân bán lẻ (POS): Tra cứu mã vạch, tính tiền, trừ kho tự động và tích điểm

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2.1.4. Luồng nghiệp vụ báo cáo thống kê doanh thu và hiệu suất ca trực

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2.2. Thiết kế mô hình kiến trúc Client - Server tổng thể

2.2.1. Sơ đồ kiến trúc 3 lớp: Desktop WinForms Client ASP.NET Core Web API SQL Server Database

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2.2.2. Sơ đồ tuần hoàn dòng chảy dữ liệu (Data Flow Diagram) trong hệ thống

2.3. Thiết kế Cơ sở dữ liệu quan hệ (Database Schema)

2.3.1. Sơ đồ thực thể liên kết (ERD) hoàn chỉnh gồm 6 bảng thực thể nghiệp vụ cốt lõi

2.3.2. Phân hệ tài khoản và phân quyền: Chi tiết bảng Users

2.3.3. Phân hệ hàng hóa và nhóm hàng: Chi tiết bảng Categories và Products

2.3.4. Phân hệ khách hàng thành viên: Chi tiết bảng Customers

2.3.5. Phân hệ hóa đơn và giao dịch bán lẻ: Chi tiết bảng Orders và OrderDetails

2.3.6. Bảng tổng hợp các ràng buộc toàn vẹn dữ liệu: Khóa chính (PK), Khóa ngoại (FK), Ràng buộc Duy nhất (Unique) và Giá trị mặc định

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2.4. Thiết kế danh mục và đặc tả chuẩn RESTful Web API

2.4.1. Quy chuẩn định dạng đường dẫn Endpoint, Header và cấu trúc gói tin JSON

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2.4.2. Phân hệ API Xác thực và Cấp phát Token (/api/auth)

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2.4.3. Phân hệ API Quản lý Nhóm hàng (/api/categories)

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2.4.4. Phân hệ API Quản lý Sản phẩm, Tra cứu Barcode và Phân trang (/api/products, /api/ado-products)

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2.4.5. Phân hệ API Giao dịch Thanh toán POS (/api/checkout)

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2.4.6. Phân hệ API Báo cáo Doanh thu Stored Procedure (/api/reports)

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CHƯƠNG 3: TRIỂN KHAI PHẦN BACKEND (ASP.NET CORE WEB API)

3.1. Cấu trúc Solution và phân tách các tầng kiến trúc

3.1.1. Cấu trúc thư mục dự án MiniSupermarket.API (Controllers, Data, Models, DTOs)

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3.1.2. Thiết lập chuỗi kết nối và thông số bảo mật an toàn trong appsettings.json

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3.1.3. Cấu hình tệp khởi động Program.cs: Đăng ký DI Container, Kích hoạt Middleware Pipeline và cấu hình CORS

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3.2. Hiện thực tầng dữ liệu với Entity Framework Core Code-First

3.2.1. Định nghĩa các thực thể C# (Model Classes) kết hợp thuộc tính Data Annotations

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3.2.2. Xây dựng lớp MiniSupermarketDbContext và cấu hình liên kết bảng qua Fluent API

3.2.3. Thực thi EF Core Migrations và khởi tạo bộ dữ liệu mẫu ban đầu (Data Seeding)

3.2.4. Triển khai các thao tác CRUD danh mục và hàng hóa qua Entity Framework Core

3.2.5. Tối ưu hóa truy vấn bảng liên quan bằng Eager Loading (.Include()) và giải quyết vòng lặp tham chiếu JSON

3.2.6. Thuật toán phân trang tầng máy chủ (Server-side Pagination) với LINQ .Skip() / .Take() kết hợp tối ưu .AsNoTracking()

3.3. Hiện thực phân hệ xác thực và phân quyền bảo mật (JWT & RBAC)

3.3.1. Xây dựng AuthController xử lý nghiệp vụ mã hóa mật khẩu và cấp phát JWT Bearer Token

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3.3.2. Cấu hình dịch vụ xác thực AddJwtBearer và thuật toán khóa ký đối xứng SymmetricSecurityKey trong Program.cs

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3.3.3. Thiết lập phân quyền truy cập đầu cuối sử dụng thuộc tính [Authorize] và [Authorize(Roles = "...")]

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3.4. Hiện thực kỹ thuật ADO.NET nâng cao và Quản trị Giao dịch POS Checkout

3.4.1. Xây dựng phân hệ tra cứu mã vạch siêu tốc bằng SqlConnection, SqlCommand và luồng đọc SqlDataReader

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3.4.2. Phòng chống triệt để lỗ hổng tấn công SQL Injection thông qua kỹ thuật tham số hóa SqlParameter

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3.4.3. Lập trình và thực thi Stored Procedure sp_GetRevenueByCashier kết xuất báo cáo doanh thu ca làm việc

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3.4.4. Đảm bảo toàn vẹn chuẩn ACID: Bọc kín chuỗi 4 thao tác thanh toán trong SqlTransaction (Commit / Rollback)

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3.5. Tài liệu hóa và chuẩn hóa kiểm thử API bằng Swagger / OpenAPI UI

3.5.1. Tích hợp thư viện Swashbuckle và cấu hình giao diện SwaggerGen

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3.5.2. Cấu hình nút xác thực ổ khóa "Authorize" để truyền Bearer Token trực tiếp trên giao diện Swagger

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CHƯƠNG 4: TRIỂN KHAI PHẦN FRONTEND CLIENT (WINDOWS FORMS POS APP)

4.1. Kiến trúc tổng thể và giao tiếp dịch vụ phía WinForms Client

4.1.1. Cấu trúc Solution dự án MiniSupermarket.WinForms

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4.1.2. Xây dựng lớp dịch vụ mạng tập trung ApiClientService.cs quản lý phiên bản static readonly HttpClient

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4.1.3. Cơ chế tự động đính kèm Bearer Token vào Header trong các yêu cầu bất đồng bộ qua System.Net.Http.Json

4.1.4. Lớp tĩnh SessionManager.cs: Quản lý phiên làm việc, lưu trữ định danh tài khoản và vai trò người dùng

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4.2. Hiện thực kiến trúc bàn làm việc Shell Workspace (Single-Form Shell)

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4.2.1. Phân chia bố cục phân vùng giao diện: Thanh điều hướng bên trái (Sidebar), Thanh tiêu đề (Top Header) và Vùng hiển thị (Main Workspace)

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4.2.2. Kỹ thuật nhúng Form con động (TopLevel = false, Dock = Fill) vào Panel trung tâm, ngăn chặn việc mở cửa sổ độc lập phân mảnh

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4.2.3. Cơ chế điều khiển phân quyền giao diện: Tự động ẩn/hiện nút chức năng Sidebar theo quyền hạn đăng nhập thực tế (RBAC)

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4.3. Xây dựng các phân hệ màn hình nghiệp vụ Desktop

4.3.1. Màn hình Xác thực đăng nhập (FormLogin.cs): Giao tiếp API xác thực, giải mã phản hồi và nạp phiên làm việc

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4.3.2. Màn hình Quầy Bán lẻ POS (FormPOS.cs): Bắt sự kiện quét mã Barcode, quản lý bảng giỏ hàng, tự động tính tổng tiền, tính tiền thừa và kích hoạt Checkout qua API

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4.3.3. Màn hình Quản lý Sản phẩm & Tồn kho (FormProductManagement.cs): Nạp danh mục ComboBox, hiển thị lưới dữ liệu DataGridView và cụm nút điều hướng phân trang (Next/Prev)

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4.3.4. Màn hình Quản lý Danh mục Nhóm hàng (FormCategoryManagement.cs): Thao tác Thêm, Sửa, Xóa và Tìm kiếm trực tiếp

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4.3.5. Màn hình Dashboard Báo cáo Doanh thu (FormReports.cs): Trực quan hóa số liệu ca trực bằng bảng biểu và điều khiển biểu đồ đồ họa

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4.4. Xử lý ngoại lệ và nâng cao trải nghiệm người dùng trên Desktop Client

4.4.1. Bọc kín các tiến trình mạng bằng cấu trúc xử lý lỗi ngoại lệ (try ... catch), chống treo hoặc sập ứng dụng

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4.4.2. Xử lý thông báo phản hồi thân thiện: Phân biệt các trạng thái lỗi mạng, hết hạn phiên làm việc (401) hoặc từ chối truy cập (403)

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CHƯƠNG 5: KIỂM THỬ HỆ THỐNG VÀ ĐÁNH GIÁ KẾT QUẢ ĐỒ ÁN

5.1. Môi trường và Kịch bản kiểm thử hệ thống

5.1.1. Cấu hình môi trường thử nghiệm: Phần cứng máy trạm, Máy chủ Web API và Cơ sở dữ liệu SQL Server

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5.1.2. Bộ dữ liệu kiểm thử chuẩn hóa: 15 tài khoản nhân sự mẫu phân quyền và 15 mặt hàng kiểm thử tồn kho

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5.2. Kiểm thử phân hệ Backend Web API (Swagger UI & Postman)

5.2.1. Kiểm thử tính năng cấp Token JWT và gọi API công khai (200 OK / 201 Created)

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5.2.2. Kiểm thử cơ chế bảo vệ phân quyền: Kiểm tra mã trạng thái 401 Unauthorized và 403 Forbidden khi truy cập vượt quyền

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5.2.3. Kiểm thử phân trang máy chủ: Đánh giá thời gian phản hồi khi trích xuất dữ liệu với các bộ tham số pageNumber và pageSize

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5.3. Kiểm thử nghiệp vụ Giao dịch toàn vẹn dữ liệu (ACID Transaction Testing)

5.3.1. Kịch bản 1 (Thanh toán thành công - Commit): Kiểm tra đối soát đồng bộ dữ liệu tại cả 4 bảng (Orders, OrderDetails, Products, Customers) trên SSMS

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5.3.2. Kịch bản 2 (Hoàn tác tự động - Rollback): Cố tình đặt mua mặt hàng vượt quá số lượng tồn kho thực tế, xác nhận hệ thống hủy giao dịch an toàn và không phát sinh hóa đơn rác

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5.4. Kiểm thử tích hợp toàn diện trên ứng dụng WinForms POS Client

5.4.1. Thử nghiệm luồng bán hàng hoàn chỉnh không dùng chuột tại quầy POS thông qua máy quét Barcode và phím tắt

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5.4.2. Thử nghiệm đồng bộ dữ liệu thời gian thực: Thao tác Thêm/Sửa/Xóa hàng hóa từ Client và kiểm tra sự thay đổi trực tiếp dưới SQL Server

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5.5. Đánh giá ưu điểm và nhược điểm của giải pháp thiết kế

5.5.1. Những ưu điểm nổi bật về mặt kiến trúc, tính bảo mật và hiệu năng vận hành

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5.5.2. Những điểm hạn chế còn tồn tại của hệ thống

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CHƯƠNG 6: KẾT LUẬN VÀ HƯỚNG PHÁT TRIỂN NÂNG CẤP

6.1. Kết quả đạt được của đồ án đối với sinh viên

6.1.1. Mức độ hoàn thành các chức năng so với yêu cầu và đề cương môn học

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6.1.2. Những kiến thức học thuật và kỹ năng kỹ thuật phần mềm tích lũy được qua đồ án (.NET Core, RESTful, EF Core, ADO.NET, WinForms)

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6.2. Hướng phát triển và nâng cấp hệ thống trong tương lai

6.2.1. Tích hợp thanh toán trực tuyến thông qua chuẩn mã VietQR động hoặc Cổng thanh toán điện tử (VNPay, MoMo)

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6.2.2. Mở rộng ứng dụng Web dành cho khách hàng tự đặt hàng (E-Commerce) và Ứng dụng Di động dành cho nhân viên kho kiểm kê hàng hóa qua camera

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6.2.3. Tích hợp thuật toán trí tuệ nhân tạo (AI/Machine Learning) dự báo nhu cầu nhập hàng và gợi ý sản phẩm bán kèm tại quầy thu ngân

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TÀI LIỆU THAM KHẢO (Sách giáo trình chuyên ngành, Tài liệu học thuật Microsoft Learn, Tiêu chuẩn mã nguồn .NET Core)

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PHỤ LỤC: HƯỚNG DẪN CÀI ĐẶT & VẬN HÀNH HỆ THỐNG

Phụ lục A: Hướng dẫn cấu hình kết nối SQL Server và các lệnh thực thi EF Core Migrations

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Phụ lục B: Danh sách bảng tài khoản người dùng mẫu và phân quyền truy cập hệ thống

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Phụ lục C: Bảng thống kê chi tiết mã nguồn các Controller và Form giao diện chính

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PHỤ LỤC (Đường link GitHub, Tài khoản/Mật khẩu demo, Hướng dẫn cài đặt nhanh)

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